

Yi Cao

AI4Science | Computer-Aided Material Design | Machine Learning | LLMs | Explainable AI | Scientific Computing

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Professional Summary

PhD researcher specializing in AI4Science with focus on machine learning applications for materials discovery. Developing novel AI frameworks that integrate large language models with first-principles simulations for accelerated materials design. Published in top-tier ML venues (NeurIPS, AAAI) with research spanning computational materials science, explainable machine learning, and causal reasoning. Strong foundation in both AI/ML methodology and scientific computing with growing interest in translating research toward practical applications.

Education

PhD	Johns Hopkins University , Chemical and Biomolecular Engineering (ChemBE)	Sept. 2023 to Present
	- GPA: 4.0/4.0 Advisor: Paulette Clancy Relevant Coursework: Machine Learning, Artificial Intelligence, Modern Data Analysis, Databases, Algorithms, Computing for Applied Mathematics, Statistical Mechanics	
BS	Fudan University , Pharmaceutical Sciences	Sept. 2019 to July 2023
	- GPA: 3.7/4.0 (Rank: 2/112) Awards: "Graduate Star" Nomination Award (2023), Excellent graduates of Shanghai Colleges and Universities (2023), 1st Class Scholarship (2021), Shanghai Scholarship (2020)	
	University of California, Berkeley , Visiting Scholar	Aug. 2021 to Dec. 2021

Research Experiences

Johns Hopkins University , Ph.D. Research under Advisor Paulette Clancy	MD, USA
- Developing AI frameworks that integrate molecular simulation (DFT, MD) with machine learning and LLMs for 2D materials discovery and design - Created dual explainability framework for machine-learned force fields (MLFFs), bridging model reasoning with human understanding (<i>XAI4Science-AAAI-2026</i>). - Developed a causal-aware KG-LLM integration framework that mitigates "contextual tunneling" and improves scientific reasoning (<i>ICLR 2026 submission</i>). - Benchmarked specialist vs. generalist MLFFs using migration pathways as diagnostic probes, revealing task-specific fine-tuning trade-offs (<i>AI4Mat-NeurIPS-2025, spotlight talk</i>). Skill set: Machine Learning (MLFFs, LLMs, causal reasoning, transfer learning), Molecular Simulation (DFT, MD), Scientific Computing	Nov. 2023 to present
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University , Undergraduate Research	Shanghai, China
- Utilized data mining and signal processing on large-scale brain recordings to uncover disease-related patterns, improving diagnostic interpretation. Skill set: Data mining, Brain-signal processing, Neuropharmacology	Feb. 2022 to Jun. 2023 16 months
School of Engineering, Westlake University , Summer Research	Hangzhou, China
Applied ML-driven polymer screening to guide nanoparticle design for brain-	July 2022 to Aug. 2022

targeted drug delivery, linking material optimization with experimental validation.

2 months

- **Skill set:** Machine Learning for materials design, Nanoparticle characterization

Technical Skills

Machine Learning & AI: Deep Learning (PyTorch, TensorFlow), Large Language Models (fine-tuning, prompt engineering), Causal Inference, Graph Neural Networks, Transfer Learning, Explainable AI

Scientific Computing: Molecular Dynamics (LAMMPS, GROMACS), Density Functional Theory (Quantum ESPRESSO), High-Performance Computing (MPI, CUDA)

Programming & Tools: Python, MATLAB, R, Git, Docker, Linux/Unix

Data Science: Large-scale data processing, Statistical analysis, Knowledge graphs, Database management (SQL), Distributed computing

Specialized: Materials informatics, Scientific AI

Publications

1. **Atomic Switch Control via Two-Mode Intercalation for Tunable 2D Materials**
Yi Cao, Victor Wu, and Paulette Clancy*
npj 2D Materials and Applications, under review (2025)
2. **What is Your Force Field Really Learning? Gaining Scientific Intuition with A Dual-Level Explainability Framework**
Yi Cao, Peter Mastracco, Jieneng Chen, Alan Yuille, Paulette Clancy*
AAAI Conference Workshop (XAI4Science), Peer-reviewed Workshop Paper (2026)  
*Dual-level explainability framework bridging model reasoning with human understanding in scientific AI
3. **When Knowledge Hurts: Enriching Domain Knowledge for Causal Scientific Reasoning**
Yi Cao, Liaoyaqi Wang, Jieneng Chen, Benjamin Van Durme, Alan Yuille, Paulette Clancy*
Preprint, submitted to ICLR Conference (2026); publicly available on OpenReview   
4. **Migration as a Probe: A Generalizable Benchmark Framework for Specialist vs. Generalist Machine-Learned Force Fields**
Yi Cao and Paulette Clancy*
NeurIPS Conference Workshop (AI4Mat), Peer-reviewed Workshop Paper (2025)   
*Comprehensive benchmarking framework for evaluating MLFF generalization in materials science
5. **Low-energy pathways lead to self-healing defects in CsPbBr₃**
Kumar Miskin, **Yi Cao**, Madaline Marland, Farhan Shaikh, David T. Moore, John Marohn, Paulette Clancy*
Phys. Chem. Chem. Phys., 27(29), 15446-15459 (2025)  
*Computational discovery of self-healing mechanisms with implications for rational material design
6. **Brain-targeted nanoparticle drug delivery systems: research advances**
Yi Cao and Chen Jiang*
Basic and Clinical Medicine, 42(1), 2–14 (2022) 

Conference Presentations

AAAI XAI4Science Workshop (Singapore Expo, Jan 2026) - **Spotlight Talk** (Top-tier recognition)

What is Your Force Field Really Learning? Gaining Scientific Intuition with A Dual-Level Explainability Framework

NeurIPS AI4Mat Workshop (San Diego, CA, Dec 2025) - **Spotlight Talk & Travel Grant Awardee** (Top-tier recognition)

Migration as a Probe: A Generalizable Benchmark Framework for Specialist vs. Generalist Machine-Learned Force Fields

Prognostics and Health Management (PHM) Society Annual Conference (Bellevue, WA, Oct 2025) - **Invited Talk & Session Chair**

Doctoral Symposium - Selected as 1 of 10 PhD students globally; showcased AI applications in predictive maintenance and materials health; Chaired "Advanced AI/ML for PHM" and "Digital Twins and Aviation" sessions

AIChE Annual Meeting (Boston, MA, Nov 2025)

Poster: Dissecting Cr-Doped Interfaces: How Diffusion Dynamics Reshape the Van Der Waals Gap and Shift Sb₂Te₃ Thermoelectric Performance

Artificial Intelligence for Materials Science (AIMS) Workshop, NIST (Rockville, MD, Jul 2025) - Poster

Women in AI 2025 (Baltimore, MD, Apr 2025) - **Poster Presentation Award Winner**

Exploring Low-Energy Pathways for Self-Healing Defects in CsPbBr₃—A Computational Study

Quantum Matters in Materials Science Workshop, NIST (Rockville, MD, Feb 2025)

Poster: Topological Materials Engineering for Next-Generation Thermoelectrics

Materials Research Society (MRS) Fall Meeting (Boston, MA, Dec 2024)

Poster: Exploring Low-Energy Pathways for Self-Healing Defects in CsPbBr₃

Additional Experiences And Awards

LLM Hackathon for Applications in Materials Science & Chemistry

Global (JHU Site, MD, USA)

Sept. 2025

Team Leader

- Led the **ARIA team** (Autonomous Reasoning Intelligence for Atomics), integrating LLMs with causal knowledge graphs for robust inverse materials design.
- Received a **2025 Visionary Award** (ranked 6th out of 120 teams).
- **Industry Recognition:** Project noticed by CuspAI, leading to ongoing collaboration discussions for commercial applications

Viva Biotech

Shanghai, China

Jun. 2024 to Jul. 2024

Internship

- Conducted Computer-Aided Drug Design (CADD) research using co-solvent MD simulations, optimizing drug discovery through protein-ligand interaction analysis.

Fudan iGEM team, iGEM Competition

Shanghai, China

Dec. 2022 to Nov. 2023

Part-time, Advisor

- Guided experimental design, scientific documentation.
- Our team won a **Gold Medal** and **Best Environmental Project** title.

Boehringer Ingelheim

Pudong, Shanghai, China

Dec. 2022 to Feb. 2023

Internship

- Led a team in developing a white paper on quality culture through research and interviews, resulting in improved company-wide quality guidelines.